

Literature Review: Piecewise Signal Estimation and Change-Point Detection

1 Trend Filtering

Trend filtering estimates an unknown regression function from

$$y_i = f_0(x_i) + \varepsilon_i, \quad i = 1, \dots, n,$$

where the design points are ordered and the errors are independent. For equally spaced inputs, the k th-order trend filtering estimator is

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^n} \left\{ \frac{1}{2} \|y - \beta\|_2^2 + \lambda \|D^{(k+1)}\beta\|_1 \right\}. \quad (1)$$

Here $D^{(k+1)}$ is the $(k+1)$ st-order discrete difference operator. The order k determines the piecewise-polynomial degree: $k = 0$ gives piecewise-constant fits, $k = 1$ piecewise-linear fits, etc. For $k = 1$,

$$\|D^{(2)}\beta\|_1 = \sum_{i=1}^{n-2} |\beta_i - 2\beta_{i+1} + \beta_{i+2}|.$$

Zero second differences correspond to locally linear fitted values, while nonzero second differences correspond to knots. Increasing λ penalizes slope changes more strongly and therefore usually reduces the number of knots.

Problem (1) is convex and belongs to the generalized-lasso family [1]. For a fixed λ , a primal-dual interior-point method exploits the sparse banded difference matrix. Each iteration costs $O(n)$, and the theoretical worst-case number of iterations is $O(\sqrt{n})$, giving approximately

$$O(n^{3/2})$$

for one fixed value of λ . A generalized-lasso path algorithm can instead compute solutions as λ varies; each path update is $O(n)$, although the number of path events need not be small.

Critique. Trend filtering is convex, has a unique fitted vector, automatically selects knot locations and has strong statistical guarantees. For $k = 1$, it directly produces continuous piecewise-linear fits. Its main drawbacks are that the number of knots is controlled only indirectly through λ , the ℓ_1 penalty shrinks slope changes and may introduce bias, and no local validity condition is imposed on fitted segments.

Experiment. A continuous piecewise-linear signal was simulated and first-order trend filtering was fitted for three values of λ .

Model	λ	Knots	MSE
Small	0.00984	12	0.00299
Medium	0.14764	2	0.00359
Large	0.78742	1	0.01467

The experiment confirms the expected regularization effect: increasing λ reduces the number of knots, while an excessively large value underfits the signal.

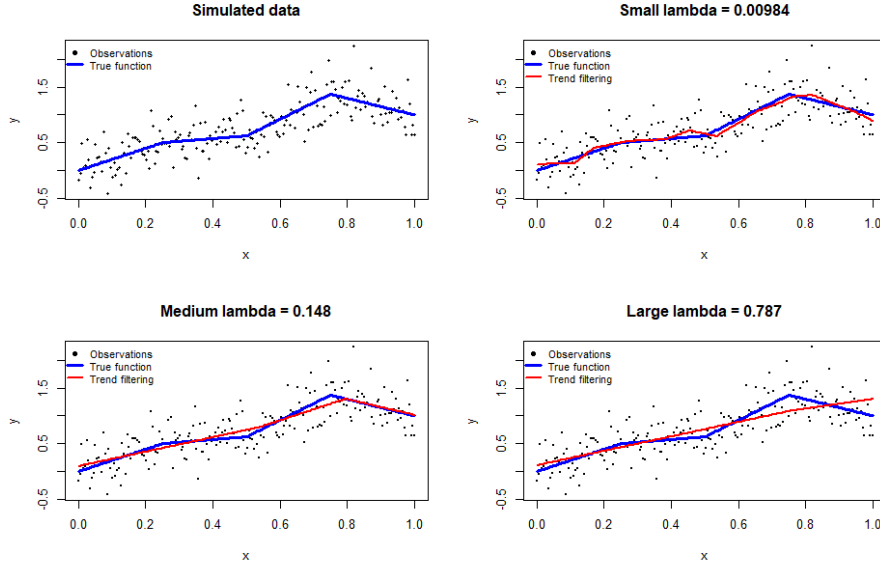


Figure 1: First-order trend filtering for several values of λ .

2 L_0 Trend Filtering

L_0 trend filtering uses the same basic regression setting but replaces the ℓ_1 penalty by a direct penalty on the number of nonzero discrete derivatives:

$$\hat{\alpha} = \arg \min_{\alpha \in \mathbb{R}^n} \left\{ \frac{1}{2} \|y - \alpha\|_2^2 + \lambda \|D^{(q+1)} \alpha\|_0 \right\}. \quad (2)$$

The quantity

$$\|z\|_0 = \#\{j : z_j \neq 0\}$$

counts nonzero components. Thus, for $q = 1$, $\|D^{(2)} \alpha\|_0$ directly counts estimated slope changes. Unlike L_1 trend filtering, the penalty does not depend on the magnitude of a change.

Problem (2) is nonconvex and combinatorial. Wen, Wang and Zhang [3] propose the AMIAS (*Alternating Minimization Induced Active Set*) method. The active set

$$A = \{j : (D^{(q+1)} \alpha)_j \neq 0\}$$

represents detected knots. Conditional on A , the primal and dual variables are obtained from banded linear systems; the active set is then updated by hard thresholding. A sequential version solves the problem for cardinalities $k = 1, \dots, k_{\max}$ using warm starts, and selects k using BIC.

Because $D^{(q+1)}$ is banded, the principal linear-algebra step costs

$$O((n - q - k - 1)q^2),$$

which is $O(n)$ per AMIAS iteration when q is fixed. This does *not* imply a global $O(n)$ solution of the original problem: AMIAS finds stationary active sets and may converge to a local solution.

Critique. The L_0 penalty directly controls the number of knots and avoids the systematic shrinkage of large changes caused by the ℓ_1 penalty. It can therefore reduce staircase effects and improve knot recovery. Its main disadvantage is nonconvexity: global optimality is not guaranteed, the result can depend on initialization, and a cardinality parameter must still be selected.

Experiment. The same piecewise-linear signal as for L_1 trend filtering was used with $q = 1$. Candidate solutions were generated for several knot counts and the final cardinality was selected by BIC.

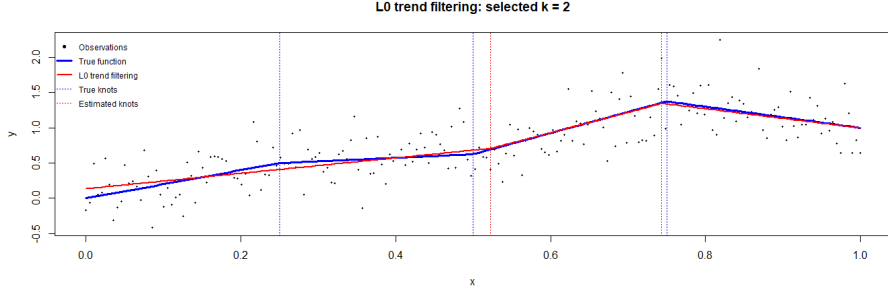


Figure 2: L_0 trend-filtering fit on the simulated piecewise-linear signal.

The experiment illustrates the main difference from L_1 trend filtering: complexity is expressed directly through the number of nonzero second differences rather than through their magnitudes.

3 Optimal Partitioning and PELT

Let

$$0 = \tau_0 < \tau_1 < \dots < \tau_m < \tau_{m+1} = n$$

be changepoints dividing a sequence into $m + 1$ segments. Optimal Partitioning (OP) minimizes

$$\sum_{i=1}^{m+1} C(y_{\tau_{i-1}+1:\tau_i}) + \beta m, \quad (3)$$

over both m and the changepoint locations. For a Gaussian change-in-mean model, a standard segment cost is

$$C(y_{a:b}) = \min_{\mu} \sum_{t=a}^b (y_t - \mu)^2.$$

Defining $F(s)$ as the minimum penalized cost for $y_1, \dots, y_s$, OP uses the recurrence

$$F(s) = \min_{0 \leq t < s} \{F(t) + C(y_{t+1:s}) + \beta\}, \quad F(0) = -\beta.$$

All possible previous changepoints are considered, so with $O(1)$ segment costs the complexity is

$$O(n^2).$$

PELT [4] accelerates OP through safe pruning. If the segment cost satisfies an appropriate subadditivity condition and

$$F(t) + C(y_{t+1:s}) + K \geq F(s),$$

then t can be proven unable to become the optimal previous changepoint at any future endpoint and is removed permanently. PELT therefore solves the *same* optimization problem as OP and retains a globally optimal solution. Its worst-case complexity remains $O(n^2)$, but under the conditions of Killick et al., especially when the expected number of changes grows linearly with n , its expected complexity is

$$O(n).$$

Critique. OP/PELT provide exact optimization for a clear global objective, and PELT can be much faster than ordinary OP. However, the number of changes is controlled through a global penalty β , and standard OP is most natural when segments can be fitted independently. Continuity constraints between successive polynomial pieces couple neighboring segments and require a more complex state formulation.

Experiment. A piecewise-constant Gaussian signal with true changepoints

$$(80, 150, 240)$$

was fitted by PELT using a Gaussian change-in-mean cost and a BIC penalty. PELT estimated the three changepoints at exactly the true locations; the MSE was 0.00446.

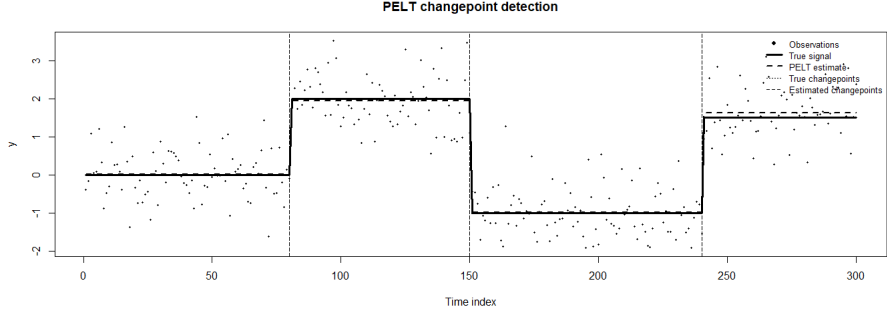


Figure 3: PELT on the simulated piecewise-constant signal.

4 Smallest Valid Partitioning

Smallest Valid Partitioning (SVP) [5] also addresses multiple change-point detection but replaces a global complexity penalty by a *local validity constraint*. For a candidate segment $y_{a..b}$, let $C(y_{a..b})$ be its fitting cost and $f(y_{a..b})$ a validity statistic. A segment is admissible only if

$$f(y_{a..b}) \leq \gamma.$$

For a segmentation with K segments, define

$$Q(\tau) = \sum_{k=0}^{K-1} C(y_{\tau_k..\tau_{k+1}}).$$

SVP minimizes the pair

$$(K, Q)$$

in lexicographic order: first minimize the number of valid segments and, among segmentations with the same K , minimize Q . Hence

$$(K, Q) \preceq (K', Q') \iff K < K' \text{ or } (K = K' \text{ and } Q \leq Q').$$

Let $R_t = (K_t, Q_t)$ be the optimal solution for the prefix ending at t . The dynamic-programming recursion is

$$R_t = \min_{\substack{s \leq t}} \{R_s + (1, C(y_{s..t})) : 0 \leq s < t, f(y_{s..t}) \leq \gamma\}, \quad R_0 = (0, 0).$$

Invalid segments never enter the optimization.

The complexity depends strongly on C and f . If both require linear time in the segment length, the total cost can reach $O(n^3)$. For many exponential-family models, C can be evaluated in $O(1)$ using cumulative statistics. If the validity statistic is γ -stable,

$$f(y_{s..t}) > \gamma \implies f(y_{s..u}) > \gamma, \quad u > t,$$

an invalid starting index can be permanently pruned. Depending on the validity test and data regime, the paper derives complexities ranging from quadratic down to linear time; with a FOCuS-updated GLR/CUSUM statistic, the general expected cost is approximately $O(n^2 \log n)$.

Critique. SVP gives explicit local control: every returned segment satisfies a chosen statistical validity criterion. It also separates the fitting cost from the validity test, allowing robust tests such as Wilcoxon or Mood. The dynamic program is exact for the SVP objective. Its limitations are the dependence on the choice and calibration of γ , potentially expensive validity computations, and possible directional effects for some sequential tests.

Experiment. For a Gaussian piecewise-constant signal with true changepoints (80, 150, 240), I used the cost-based validity statistic

$$f_{\text{OP}}(y_{a..b}) = \max_{a < u < b} [C(y_{a..b}) - C(y_{a..u}) - C(y_{u..b})]$$

with

$$\gamma = 2 \log(n) = 11.4076.$$

SVP returned three changepoints exactly at (80, 150, 240) and achieved MSE 0.001114.

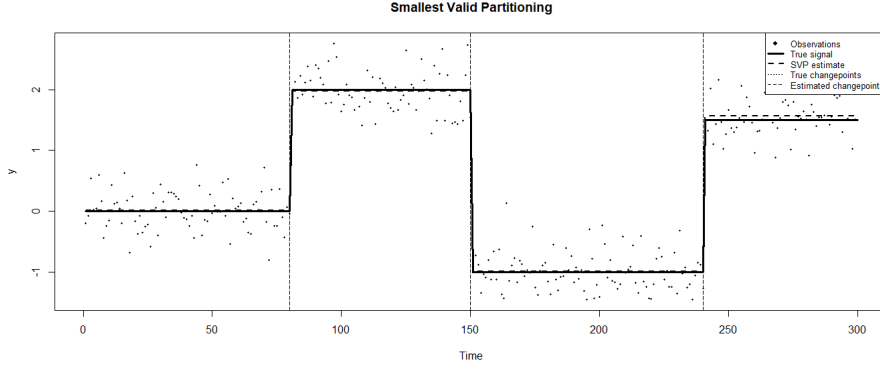


Figure 4: SVP on the simulated piecewise-constant Gaussian signal.

5 Comparison

Method	Complexity control	Algorithm	Complexity	Main limitation
L_1 -TF	ℓ_1 penalty on slope changes	Convex generalized lasso	$O(n^{3/2})$ for fixed λ	Shrinkage bias; number of knots controlled indirectly
L_0 -TF	L_0 penalty on the number of slope changes	AMIAS active-set search	$O(n)$ per iteration for fixed q	Nonconvex; no global guarantee
OP / PELT	Global penalty βm on the number of change-points	Exact dynamic programming with pruning	OP: $O(n^2)$; PELT: expected $O(n)$ under assumptions	No local segment-validity requirement
SVP	Local validity threshold γ ; minimizes number of valid segments	Lexicographic dynamic programming	$O(n)-O(n^3)$ depending on validity test and regime	Depends strongly on validity rule and threshold calibration

Table 1: Summary of the four reviewed approaches.

The methods differ mainly in how complexity is controlled. Trend filtering penalizes derivative magnitudes, L_0 trend filtering penalizes their number, OP/PELT uses a global changepoint penalty, while SVP imposes local segment validity and then minimizes the number of valid segments. For the planned continuous piecewise-linear SVP model, the main open difficulty is to combine the local-validity principle of SVP with continuity constraints between adjacent fitted lines.

References

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